

EGGROLL Accelerated Gradient Descent

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EGGROLL update:

$$\tilde{g}(x) = \frac{1}{r} \sum_{i=1}^r \frac{f(x + \sigma p_i) - f(x - \sigma p_i)}{2\sigma} p_i$$

Nesterov AGD with parameter θ :

$$\begin{aligned} x_t &= y_{t-1} - \eta \tilde{g}(y_{t-1}) && \text{(progress)} \\ y_t &= x_t + (1 - \theta)(x_t - x_{t-1}) && \text{(momentum)} \end{aligned}$$

Define Hamiltonian

$$E_t = \sqrt{\eta} f(x_t) + \frac{1}{2\sqrt{\eta}} \|v_t\|^2$$

where $v_t = x_t - x_{t-1}$ is the velocity at time t .

It follows that

$$\begin{aligned} f(x_{t+1}) + \frac{1}{2\eta} \|v_{t+1}\|^2 &\leq f(x_t) + \langle \nabla f(y_t), y_t - x_t \rangle - \eta \langle \nabla f(y_t), \tilde{g}(y_t) \rangle \\ &\quad + \frac{L\eta^2}{2} \|\tilde{g}(y_t)\|^2 + \frac{(1 - \theta)^2}{\eta} \|x_t - x_{t-1}\|^2 \\ &\quad - \langle \tilde{g}(y_t), y_t - x_t \rangle + \frac{\eta}{2} \|\tilde{g}(y_t)\|^2. \end{aligned}$$

The term we care most about is

$$\nabla f(y_t)^\top (I - \theta P P^\top)(y_t - x_t)$$

Run EGGROLL with $\tau = t_f(\delta)$ many steps. Then with probability at least $1 - \delta$, for any starting point x_0 , we have

$$f(x_\tau) - f(x_0) \leq -\frac{\eta}{4} \min_{0 \leq t \leq t_f(\delta)} \|\nabla f(X_t)\|^2 \leq -\frac{\eta}{4} \|\nabla f(X_0)\|^2$$

Now, define $h := x_\tau - x_0$, we will show this has good alignment with the gradient. Observe:

$$\|x_\tau - x_0\| = \eta \left\| \sum_{t=1}^{\tau} \frac{1}{r} \sum_{i=1}^r \langle \nabla f(x_i), P_{t,j} \rangle P_{t,j} \right\| \leq 4\tau\eta \sqrt{\frac{dm}{r}} \|\nabla f(x_0)\|$$

Therefore,

$$\frac{\langle \nabla f(x_0), -h \rangle}{\|\nabla f(x_0)\| \|h\|} \geq \frac{1}{16\tau} \sqrt{\frac{r}{dm}}$$

Meaning, $-h$ is aligned decently with the gradient, and stepping towards $-h$ decreases function value.